THE CHINESE ECONOMY

A JOURNEY INTO HISTORY, ECONOMICS, AND POLICY

Lecture 4: From the farm to the factory. The hukou

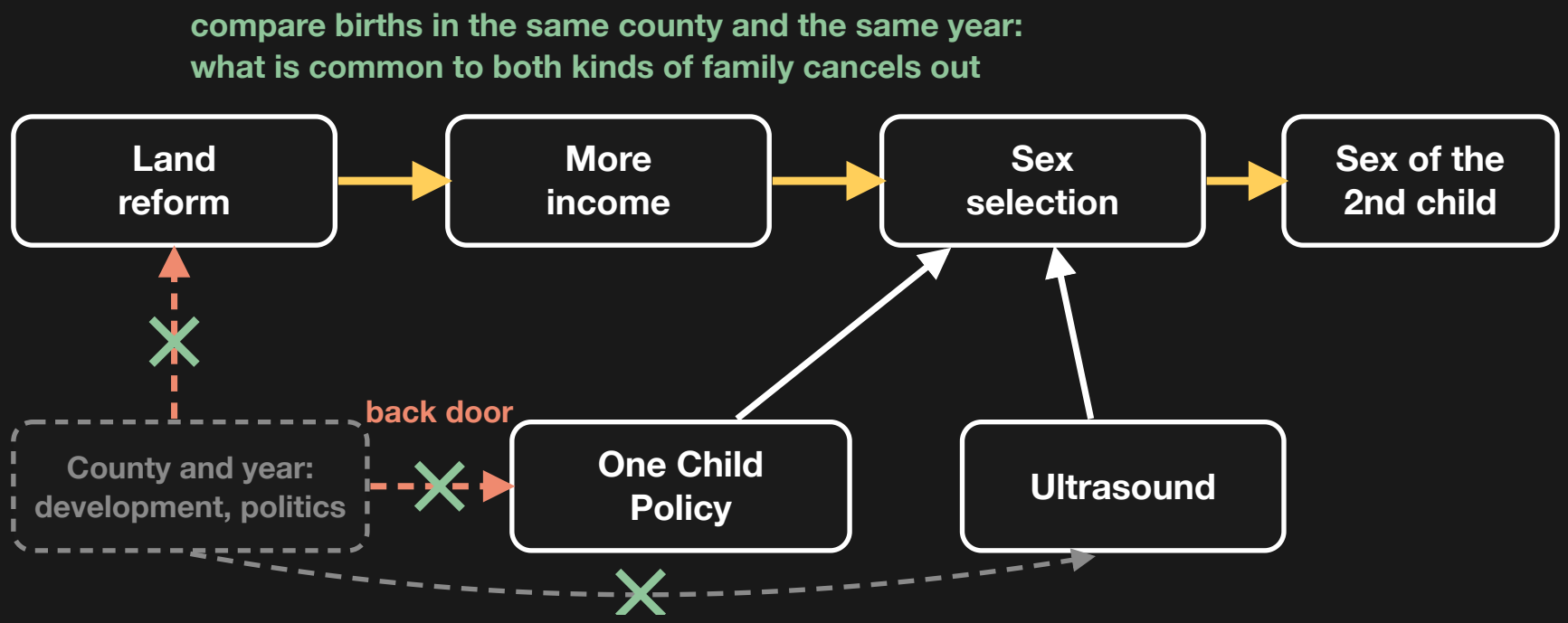
TODAY

- **Your answers**: two papers, two causal graphs, one question each that leads us on
- **The hukou**: a booklet that decided where you could live and what you could eat
- **Two sectors**: why moving workers from the farm to the city raises output, with no new technology
- **Homework** for tomorrow

QUESTION 1

ALMOND, LI AND ZHANG (2019) · LAND REFORM AND THE MISSING GIRLS

Draw the causal graph: land reform, income, sex selection, the sex of the second child. Add the One Child Policy and ultrasound. Which arrows create a "back door"?



QUESTION 2

ALMOND, LI AND ZHANG (2019) · LAND REFORM AND THE MISSING GIRLS

Why are first births and second births after a boy so useful? Which rival explanations do they rule out, and which not?

- **They rule out** anything that hits **every** family in the county that year: a harvest, a clinic, a campaign moves all the rows
- **They cannot rule out** something that arrives with the reform and matters **only** to families who want a son. Example: **ultrasound**
- That is why the authors look at **rail links** to the provincial capital, where the machines were

QUESTION 4

ALMOND, LI AND ZHANG (2019) · LAND REFORM AND THE MISSING GIRLS

State in one sentence the assumption that makes the comparison causal.

- Without the reform, the gap in the share of boys between "after a girl" and "after a boy" families would have **stayed the same**
- We cannot see it after the reform. We can check it **before**: year by year, the two lines run **flat and together** until year 0
- Same idea as lecture 3: **parallel trends**, an assumption, not a fact

QUESTION 5

ALMOND, LI AND ZHANG (2019) · LAND REFORM AND THE MISSING GIRLS

Describe the ideal experiment to learn whether higher income raises sex selection. Why could no one run it?

- Take many rural families, give extra income to some **chosen by lottery**, and record the sex of their next children
- Why not: decades of births, millions of families, and an experiment whose outcome is **girls who are not born**
- What the paper uses instead: **nature's lottery** (the sex of the first child) and the **timing** of a reform decided by others

QUESTION 6

ALMOND, LI AND ZHANG (2019) · LAND REFORM AND THE MISSING GIRLS

Explain in your words why the sex ratio of first births does not change after the reform.

- **Nature decides** the first child: about 1.05 boys per girl, before and after
- Families select only when they **already have a girl** and want a son: the reason to select comes with the second birth
- If the reform had made sons more **useful** on the farm, families would select already at the first birth. They do not: the **income** story fits better

QUESTION 9

ALMOND, LI AND ZHANG (2019) · LAND REFORM AND THE MISSING GIRLS

The census does not ask about birth order: the authors rebuild it inside each household. Which families could be misclassified? Could unregistered girls look like "missing girls"? Which way would that push the result?

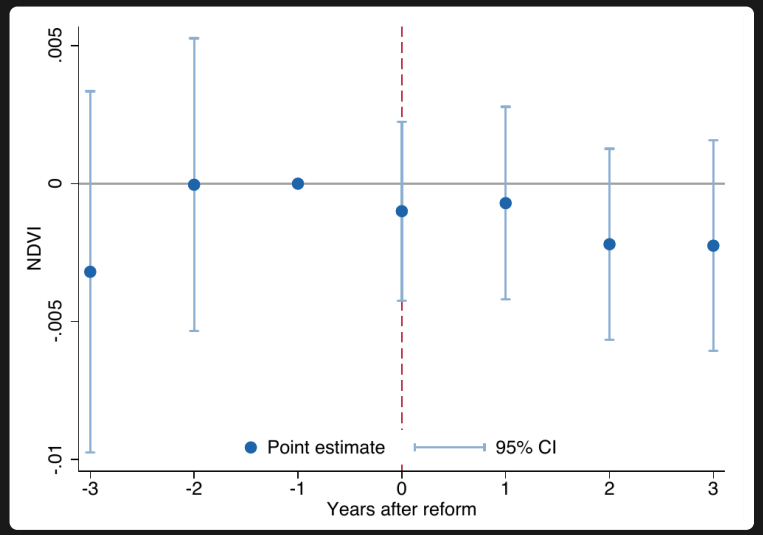
- Misclassified: a first child who **died**, who **lives elsewhere**, who was **given away**; a girl who was **never registered**
- If hidden girls became more common **after a first girl** and **after the reform**, the "missing girls" would be partly **hidden girls**: the effect would look too large
- The habit to keep: for each problem in the data, ask **which direction** it pushes the result

QUESTION 13

ALMOND, LI AND ZHANG (2019) · LAND REFORM AND THE MISSING GIRLS

Almond, Li and Zhang find that grain output per person rose after the reform; Ferguson and Kim find no effect on yields seen from space. Can both be right?

- **Ferguson and Kim** (the paper we skipped): fields seen from **satellites**, compared across provincial borders before and after. **No effect on yields**
- Different **measures**: output per person, or output per field
- Different **sources**: a statistical office, or a satellite
- Their reading: **prices** rose in 1979, and the reform **freed labour** for other work. Where did it go? **Today's topic**

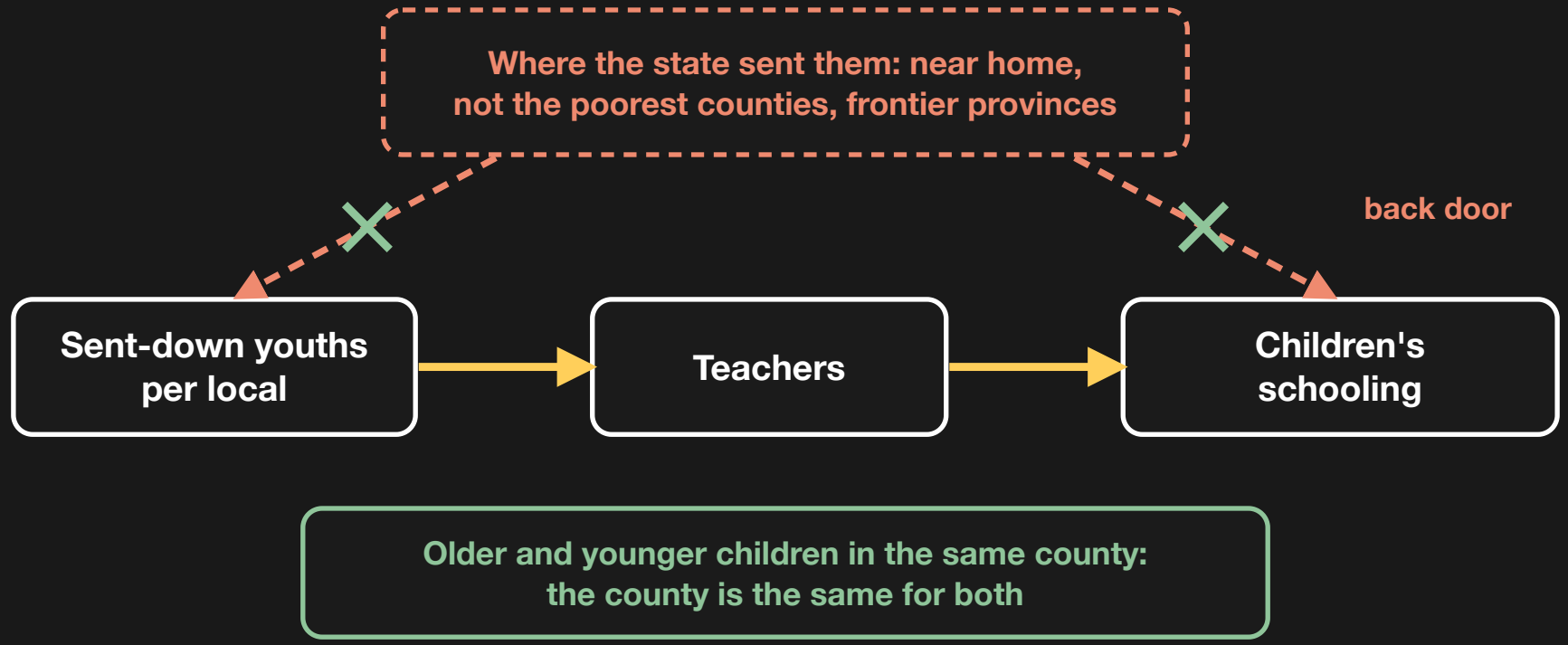


Ferguson and Kim (2026), Fig. 6: change in the border gap in vegetation after a province reforms, years 0-3

QUESTION 1

CHEN, FAN, GU AND ZHOU (2020) · CITY YOUTHS IN VILLAGE SCHOOLS

Draw the causal graph: density of sent-down youths, number of teachers, children's schooling. Add the government's choice of destinations. Where is the back door?

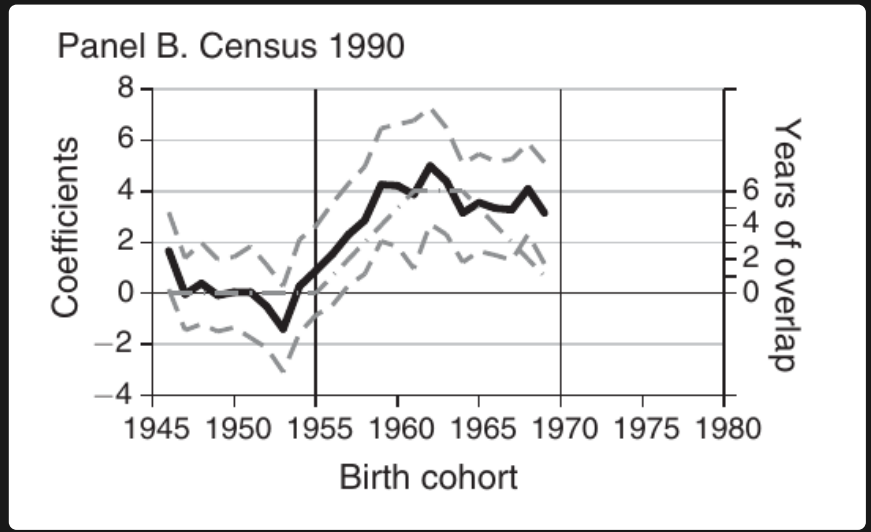


QUESTION 2

CHEN, FAN, GU AND ZHOU (2020) · CITY YOUTHS IN VILLAGE SCHOOLS

The authors say their method does not need youths to be sent at random. Explain why, using the graph and the two differences.

- **First difference**: across counties, many youths or few
- **Second difference**: across birth years. Born 1956-69, of school age while the youths were there; born 1946-55, too old
- Whatever is special about a county hits **old and young alike**, and cancels
- The assumption: without the youths, schooling would have risen from older to younger cohorts at the **same pace** everywhere. Flat before 1955



Chen, Fan, Gu and Zhou (2020), Fig. 3B: link between youth density and schooling, by birth cohort (1990 census)

QUESTION 7

CHEN, FAN, GU AND ZHOU (2020) · CITY YOUTHS IN VILLAGE SCHOOLS

Why is 0.07 years per child both a small number and a large one?

- About **0.07 years** of schooling per rural child, in the average county
- **Small**: less than a month of school per child
- **Large**: an average over **all** children; 245 million children make **17.6 million** extra years; every 100 youths, about **21** more primary and **36** more junior high graduates
- And a **lower bound**: older cohorts gained a little too, and the most educated had often left by 1990

QUESTION 8

CHEN, FAN, GU AND ZHOU (2020) · CITY YOUTHS IN VILLAGE SCHOOLS

Girls gained about twice as much as boys. The authors suggest that educated young women acted as role models. How would you test this idea with data?

- The fact: girls about **+0.09** years, boys about **+0.06**
- A test: compare counties that received more young **women** with counties that received more young **men**. If role models matter, girls gain more where the women went
- Other traces to look for: later **views on education**, jobs, age at marriage, by sex
- And the rival to rule out: girls simply had **more to gain**, starting from less schooling (4.2 years against 6.5)

QUESTION 14

CHEN, FAN, GU AND ZHOU (2020) · CITY YOUTHS IN VILLAGE SCHOOLS

How could this episode have helped China's growth after 1978? Think of who worked in township enterprises and coastal factories in the 1990s.

- The authors' answer: the rural workers who moved to **factories and township enterprises** in the 1990s had more schooling, partly because of the youths
- Moved: from the farm to the factory, from the village to the city
- But in China, moving was **not free**. A teacher could arrive in the village only because the state **sent** her. A farmer could not simply go
- Why? **The hukou**

A BOOKLET THAT DECIDES WHERE YOU LIVE



Household Register (居民户口簿), cover of the 1996 design. Wikimedia Commons, public domain

- Every household has one. Each person is registered with two things:
- a **category**: agricultural or non-agricultural
- a **place**: the village, town or city where you belong
- Both are **inherited**, in practice from the mother. You could move down the ladder, hardly up
- Working outside your place or category: **prohibited**

INSIDE THE BOOKLET



注 意 事 项

一、居民户口簿具有证明公民身份状况以及家庭成员间相互关系的法律效力，是户口登记机关进行户籍调查、核对的主要依据。户口登记机关工作人员进行户籍调查、核对时，户主或本户成员应主动交验居民户口簿。
二、户主对居民户口簿应妥善保管，严禁私自涂改、转让、出借。如有遗失，须立即报告户口登记机关。
三、居民户口簿登记权属于户口登记机关，其他任何单位和个人不得在簿上作任何记载。
四、本户如有人员增减或者登记事项发生变动，应持居民户口簿到户口登记机关申报登记。
五、全户迁出户口管辖区的，应向户口登记机关缴销居民户口簿。

户别 非农业家庭户口 户主姓名
户号 住址 本溪市

辽宁省公安厅 户口专用章
省级公安机关户口专用章
已验
户口登记机关户口专用章
辽宁省本溪市 区 派出所

承办人签章： 2008 年 07 月 16 日 签发

登记事项变更和更正记载

项目	变更、更正后	变动日期	承办人签章

非农业家庭户口 常住人口登记卡

姓名		户主或与户主关系	长子
曾用名		性别	男
出生地	辽宁省本溪市	民族	汉族
籍贯	辽宁省大连市	出生日期	年 月 日
本市(县)其他住址		宗教信仰	无宗教信仰
公民身份证件编号	2105	身高	血型 不明
文化程度	上初中	婚姻状况 未婚	兵役状况 未服兵役
服务处所	无服务处所		职业 学生
何时由何地迁来本市(县)			
何时由何地迁来本址			

刘亚茹 2008 07 16
承办人签章： 登记日期： 年 月 日

Inside pages of a booklet issued in 2008 (Benxi, Liaoning), personal data blurred. Wikimedia Commons, public domain

- 户别 **type of household**: 非农业家庭户口, "non-agricultural household". The **category**
- 住址 **address**: Benxi city. The **place**
- 何时由何地迁来 **"when and from where moved here"**: every move goes through the police station that keeps the register

WHY: FEEDING THE CITIES CHEAPLY

- **1953**: the state buys grain from farmers in fixed amounts, at **low prices**, to feed the cities and fund industry
- **1955**: grain **rationing** in the cities (for over 30 years); 1956, families classified as agricultural or not
- **1958**: the household registration regulation. To move to a city you need a **job** or **school** certificate, or a permit from the destination's **police**
- After the Great Leap: urban registration is a **"meal ticket"**, rural registration **"a license to starve"**
- 1961-62: **six million** recent arrivals sent back to the villages. By the mid-1960s, a **"caste system"**

Naughton (2007), pp. 115-116; Cheng and Selden (1994); Tombe and Zhu (2019), p. 6

CHANGING STATUS: THE STATE DECIDES

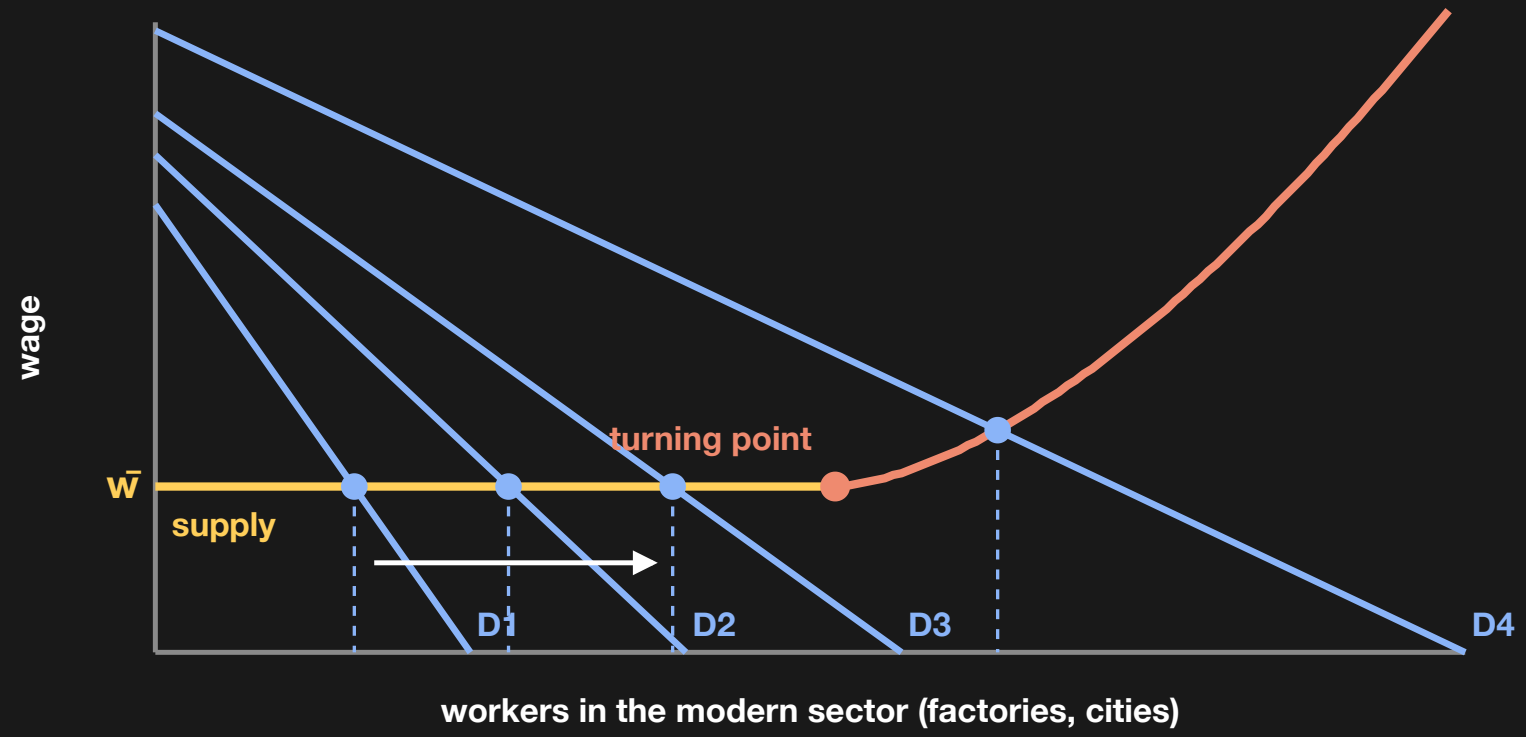
- Moving from rural to urban status had a name: ***nongzhuanfei***, "from agricultural to non-agricultural"
- Who got it: **permanent recruits** of state enterprises, people whose land the state **took**, students admitted to **university**, people promoted to **official posts**, demobilised **soldiers**
- How many: a yearly **quota** set in Beijing, **0.15-0.2%** of each place's non-agricultural population
- Selection by the **state's needs**, not by the worker's wish or by the market. Keep this in mind for next week: who is selected, and by what criterion, will change

Chan and Buckingham (2008), pp. 590-591

THE QUESTION

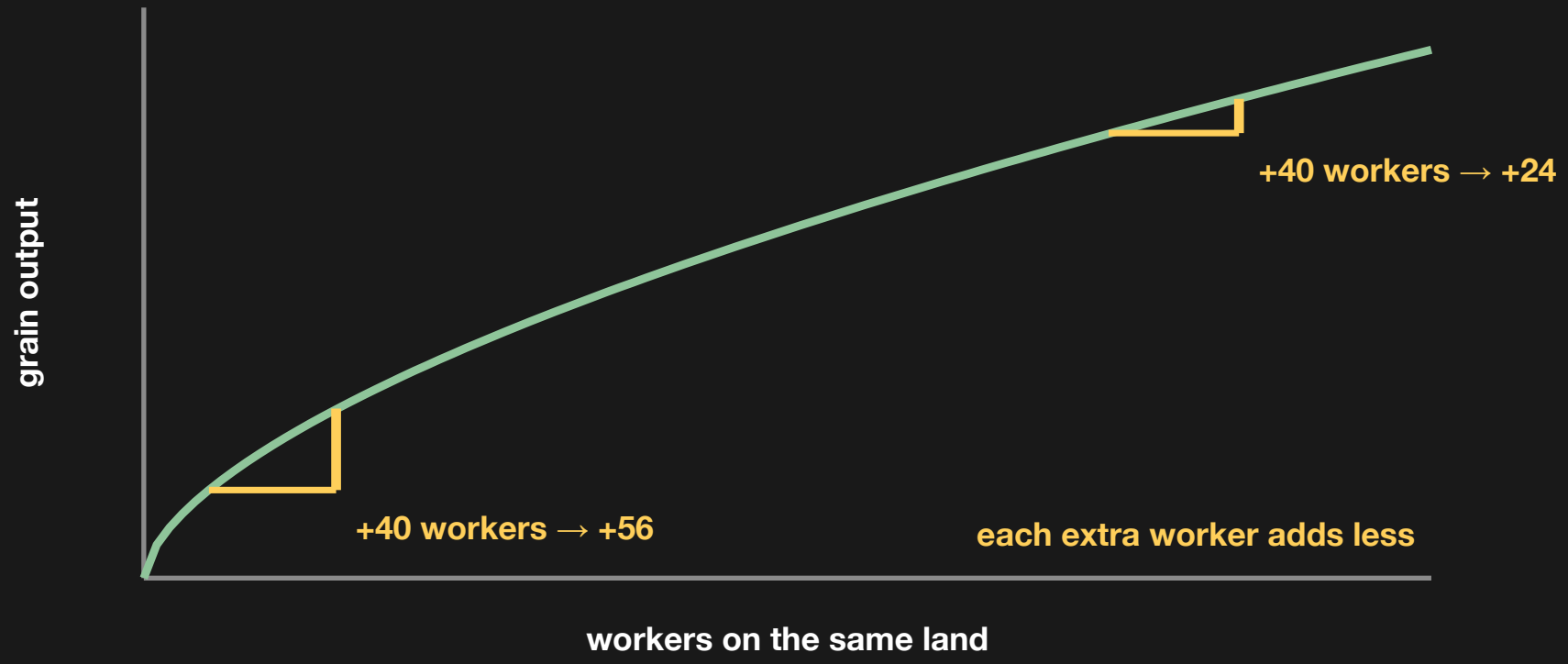
- A job in a city factory paid more, and came with a ration card, a house, a pension
- Most people could not take it. In 1978 about **four in five** Chinese lived in the countryside
- **If workers cannot move, how much does the economy lose? And how would we know?**

LEWIS (1954): TWO ECONOMIES IN ONE



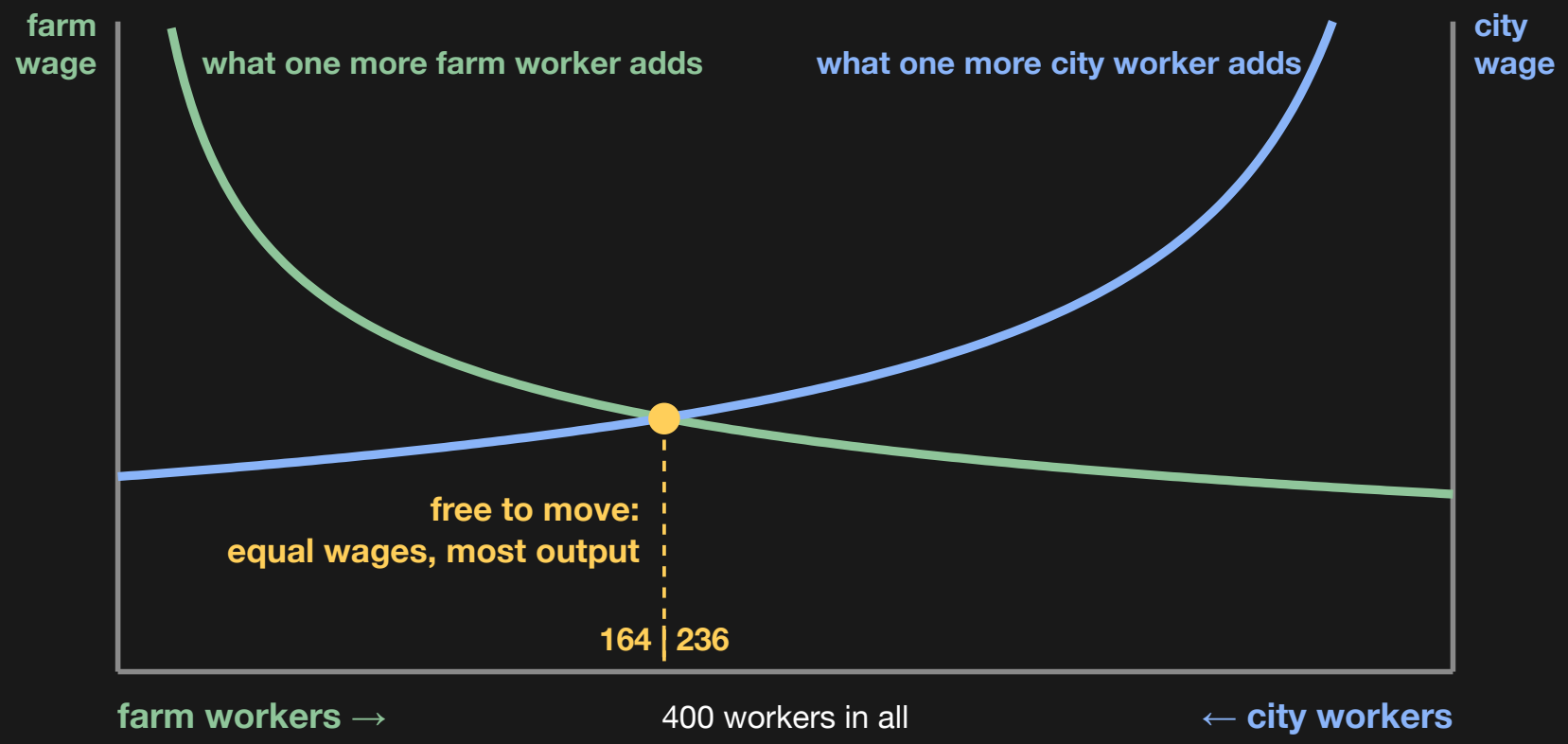
- A **traditional** sector with too many hands: one worker fewer costs almost no output. Factories can hire at a **fixed wage** w̄
- Factories hire until one more worker adds no more than w̄ (D1)
- Profits are reinvested, factories grow, they hire more **at the same wage**: growth is moving people
- When the extra hands run out: the **turning point**. Wages start to rise (lecture 5)

ONE MORE WORKER ON THE SAME LAND



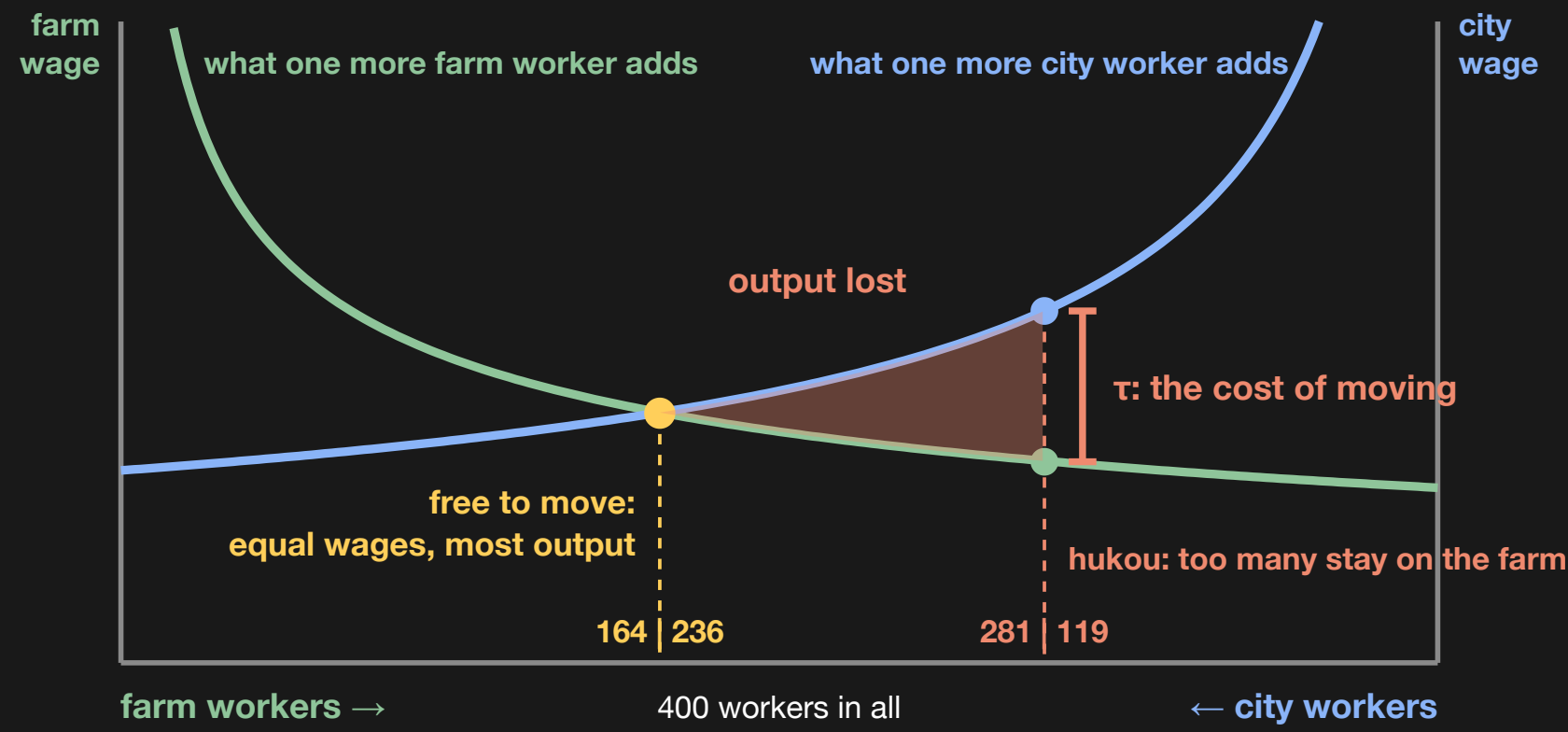
- A **production function**: how much output from how many workers, with the same land and tools
- The land is fixed: each extra worker adds **less**. What one more worker adds is the **marginal product**; with prices, it is what a worker is worth to an employer

TWO SECTORS, ONE LABOUR FORCE



- 400 workers: farm workers counted **from the left**, city workers **from the right**
- Free to move, workers go where they earn more, until wages are **equal**. There, no move can raise total output

ADD THE HUKOU: A COST OF MOVING



- Moving now costs **τ**: no ration, no school, no pension, the land left behind. A worker moves only if the city pays at least τ more
- Too many stay on the farm. Each one held back would have added more in the city than on the farm: the red area is **output lost**

TWO WARNINGS

- **A wage gap is not a welfare gap.** The last worker who stays is **indifferent**: the city pays τ more, and moving costs τ. The gap tells us about the **barrier**, not about who is worse off
- **τ is a cost, not a tax.** Nobody collects it. Lost rations, lost land, lost schooling: no one gets them. So the output is **lost**, not moved to someone else
- The first worker held back costs little; each extra one costs more. A **small** barrier is cheap, a **large** one very expensive

LEWIS AND THE WEDGE

	Lewis (1954)	Our model, with the hukou
One farm worker fewer	costs almost nothing	costs a little, more and more
Wages	flat until the turning point	equal if free; apart by τ with the hukou
Moving people	raises output	raises output, until wages are equal
What we can measure	when the extra hands run out	τ, from wages and flows of migrants

TRY IT: THE HUKOU WEDGE LAB



Open the lab

- Move the slider: how many workers stay on the farm, how far apart wages go, how much output is lost
- Press **"Relax the hukou"**: what happens to the gap, and to output?

NEXT TIME: τ IS A NUMBER

- τ is not just a letter: you can **estimate** it from how many people move and how much wages differ
- Tombe and Zhu did, for China in 2000: moving from the countryside to a city **in the same province** cost as much as cutting your real income by a factor of **nearly 3**
- Then it **fell**. Next time: the reforms, what they did to China's growth, and who was let in
- **Think about**: city wages were higher for decades, yet many did not move. What does the model say about why, and what does it **not** say?

HOMEWORK FOR TOMORROW

1. Back to last week's questions

- Answer the questions we did **not** discuss today:
  Almond, Li and Zhang: 3, 7, 8, 10, 11, 12, 14
  Chen, Fan, Gu and Zhou: 3, 4, 5, 6, 9, 10, 11, 12, 13, 15
- Go over the answers we gave today, until you could **explain them** to someone else

2. Play with the Hukou Wedge Lab

Then try to answer:

1. Compare the farm wage and the city wage with and without the hukou. **Who gains** from the hukou, and who loses? Who might want to **keep** it?
2. Press "Urban sector much more productive". With the same hukou, are more workers held back? Is more output lost? What does it say about the cost of the hukou as China's cities grew richer?
3. In Lewis's picture, one farm worker fewer costs **almost nothing**. Would the hukou cost more or less output than in our model? Why?